

# SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

## BizMind BD

### AI-Powered SME Commerce, Procurement & Business Intelligence Platform

*Web Programming and Mobile Application Development | Version 1.0 | August 2026*

#### 1. Introduction

BizMind BD is a proposed web and Android platform for small and medium-sized enterprises (SMEs) in Bangladesh. It combines business management, B2B commerce, procurement, supplier management and business intelligence with an Agentic AI layer.

The project is designed as a version-based product. Version 1.0 will establish the core business foundation and an Agentic AI prototype. Future versions can add advanced forecasting, supplier networks, logistics, financing support and industry-specific capabilities without redesigning the entire system.

#### 2. Purpose

This SRS defines the initial scope, objectives, users, functional and non-functional requirements, AI capabilities, system boundaries and planned evolution of BizMind BD. It will be the baseline for design and development.

#### 3. Problem Statement

Many SMEs in Bangladesh manage sales, inventory, procurement, suppliers and expenses through manual records, spreadsheets, phone calls or fragmented applications. This makes it difficult to obtain a unified view of the business, predict demand, compare suppliers objectively, identify risks and make timely purchasing decisions.

Basic management systems mainly record transactions. BizMind BD aims to go further by analyzing business information and assisting owners with decisions and approved operational tasks.

#### 4. Proposed Solution

BizMind BD will provide a centralized Web and Android platform for products, sales, inventory, customers, suppliers, purchases and expenses. A business-intelligence layer will produce dashboards and insights. An Agentic AI layer will use specialized agents and approved tools to analyze problems, forecast demand, compare suppliers, recommend actions and prepare selected actions for owner approval.

High-impact actions will remain under human control; the AI will not have unrestricted authority.

#### 5. Scope

##### 5.1 In Scope — Version 1.0

- Authentication and role-based access
- Business and employee management
- Product catalog and inventory
- Sales and customer management
- Supplier management
- B2B supplier/product listings and comparison
- Purchase requests and purchase orders
- Expense and basic financial tracking
- Business KPI dashboard
- Low-stock and business notifications
- Basic demand forecasting
- Supplier scoring/recommendation
- Agentic AI business insights
- AI-assisted procurement recommendation

- Human approval workflow
- Web application and essential Android functions

## 5.2 Out of Scope — Version 1.0

- Direct banking/payment processing
- Automatic financial transactions without approval
- Full accounting, tax filing or legally certified reports
- Nationwide logistics infrastructure
- Production-scale lending/credit scoring
- Training an LLM from scratch

## 5.3 Version Roadmap

| Version | Focus              | Major Additions                                                            | Goal                               |
|---------|--------------------|----------------------------------------------------------------------------|------------------------------------|
| V1.0    | Foundation         | Core SME management, procurement, dashboards, basic Agentic AI             | Working academic/startup prototype |
| V1.1    | Intelligence       | Improved forecasting, anomaly detection, advanced KPIs                     | Better business insights           |
| V2.0    | Commerce Network   | Multi-business supplier network, quotation comparison, order tracking      | B2B ecosystem                      |
| V2.1    | Automation         | Auto-replenishment suggestions, recurring procurement, workflow automation | Reduce routine work                |
| V3.0    | Business Ecosystem | Logistics integration, financing partnerships, industry-specific agents    | Scalable startup platform          |

## 6. Objectives

- Build an integrated SME platform for the Bangladesh context
- Reduce manual work in inventory and procurement
- Support data-driven business decisions
- Forecast product demand
- Recommend suitable suppliers
- Demonstrate meaningful Agentic AI rather than only chatbot functionality
- Provide controlled AI-assisted automation
- Use GitHub-based collaborative development
- Maintain an architecture that supports future versions

## 7. Stakeholders and User Roles

| Role                   | Responsibilities                                                         | Access       |
|------------------------|--------------------------------------------------------------------------|--------------|
| Business Owner         | Manage business, monitor performance, approve AI recommendations/actions | Web + Mobile |
| Employee               | Record sales, update stock and perform assigned tasks                    | Web + Mobile |
| Supplier               | Manage products, prices, orders and delivery information                 | Web          |
| Platform Administrator | Manage platform users, businesses, suppliers and monitoring              | Web          |
| AI Agent System        | Analyze data, call approved tools and                                    | Internal     |

|  |                         |  |
|--|-------------------------|--|
|  | prepare recommendations |  |
|--|-------------------------|--|

## 8. Functional Requirements

| ID    | Requirement            | Description                                                             |
|-------|------------------------|-------------------------------------------------------------------------|
| FR-01 | Authentication         | Registration, login, logout, password management and role-based access. |
| FR-02 | Business Management    | Owners manage business profiles and employees.                          |
| FR-03 | Product Management     | Authorized users add, edit, categorize, search and deactivate products. |
| FR-04 | Inventory              | Track quantities, stock movements, adjustments and low-stock status.    |
| FR-05 | Sales                  | Record sales and update inventory.                                      |
| FR-06 | Customers              | Maintain basic customer information and transaction history.            |
| FR-07 | Suppliers              | Maintain supplier profiles, prices and performance information.         |
| FR-08 | B2B Commerce           | Browse supplier/product listings and compare purchasing options.        |
| FR-09 | Procurement            | Create purchase requests/orders and track status.                       |
| FR-10 | Expenses               | Record and categorize business expenses.                                |
| FR-11 | Business Intelligence  | Show sales, inventory, purchasing, expense and KPI dashboards.          |
| FR-12 | Notifications          | Provide stock, order, business and AI alerts.                           |
| FR-13 | Demand Forecasting     | Estimate future product demand from historical sales.                   |
| FR-14 | Supplier Intelligence  | Score suppliers using price, reliability, delivery and history.         |
| FR-15 | Business Analyst Agent | Identify selected trends, anomalies, risks and opportunities.           |
| FR-16 | Procurement Agent      | Recommend what, how much and from whom to purchase.                     |
| FR-17 | AI Assistant           | Allow owners to ask business questions in natural language.             |
| FR-18 | Agent Orchestration    | Route tasks to specialized agents and approved tools.                   |
| FR-19 | Human Approval         | Require explicit approval for selected AI-generated actions.            |
| FR-20 | Mobile Application     | Support essential business operations, notifications and AI monitoring. |
| FR-21 | Audit Trail            | Record important user and AI actions for traceability.                  |

## 9. Agentic AI Requirements

The defining feature is an agent-based AI layer. An Agent Orchestrator will identify the business problem, invoke specialized agents and approved tools, combine results and prepare an actionable recommendation.

| Agent                       | Function                 | Example                                         |
|-----------------------------|--------------------------|-------------------------------------------------|
| Agent Orchestrator          | Coordinates agents/tools | Selects demand, supplier and procurement agents |
| Demand Forecasting Agent    | Forecasts demand         | Predicts next 7/14/30 days                      |
| Procurement Agent           | Plans purchasing         | Calculates recommended quantity                 |
| Supplier Intelligence Agent | Evaluates suppliers      | Compares price, delivery and reliability        |

|                        |                               |                                              |
|------------------------|-------------------------------|----------------------------------------------|
| Business Analyst Agent | Analyzes performance          | Detects declining sales/unusual expenses     |
| Finance Agent          | Monitors financial indicators | Highlights expense growth/cash-flow pressure |

### 9.1 Example Workflow

Owner asks: “Do I need to purchase more cooking oil?” The system checks inventory, retrieves sales history, forecasts near-term demand, calculates required quantity, compares suppliers and generates a recommendation. The owner reviews and approves the proposed purchase order, after which the system creates the approved order record.

## 10. Non-Functional Requirements

| Category        | Requirement                                                                         |
|-----------------|-------------------------------------------------------------------------------------|
| Security        | Protect passwords/credentials and enforce role-based authorization.                 |
| Usability       | Be understandable to SME users with limited technical knowledge.                    |
| Performance     | Common operations should respond within an acceptable time under expected load.     |
| Scalability     | Support additional businesses, products, suppliers, users and AI agents.            |
| Reliability     | Maintain consistency of sales, inventory and procurement transactions.              |
| Maintainability | Keep modules separated and documented for future versions.                          |
| Privacy         | Restrict business data to authorized users.                                         |
| Explainability  | Provide understandable reasons/supporting factors for important AI recommendations. |
| Auditability    | Keep traceable records of important user and AI-generated actions.                  |

## 11. Technology Stack

Web: React.js/Next.js | Mobile: Kotlin + Android Studio | Backend: Node.js/Express.js | AI/ML: Python/FastAPI, Scikit-learn/XGBoost | Database: PostgreSQL | Agentic AI: LangGraph/LangChain + an LLM API such as OpenAI or Gemini | Communication: REST/JSON | Notifications: Firebase Cloud Messaging | Version control: Git/GitHub.

## 12. High-Level System Workflow

Web/Mobile User → Authentication → Backend/API → Business Database → Analytics/ML → Agent Orchestrator → Specialized Agents → Approved Tools/Data → Recommendation → Human Approval → Business Action → Audit/Notification.

## 13. Initial Data Entities

- User
- Business
- Role/Employee
- Product
- Category
- Inventory
- Inventory Transaction
- Sale
- Sale Item
- Customer
- Supplier
- Supplier Offer
- Purchase Order

- Purchase Order Item
- Expense
- Notification
- AI Recommendation
- AI Action/Audit Log

A detailed ER diagram and normalized database design will be produced in Week 2.

#### 14. UI and Platform Requirements

- Responsive web dashboard
- Android mobile interface for essential operations
- Business KPI cards and charts
- Search/filter for products, suppliers, sales and orders
- Natural-language AI assistant
- Visible explanation and approval controls for AI recommendations
- Notification center

#### 15. Development Plan

| Phase           | Activities                                                                                   | Expected Outcome                     |
|-----------------|----------------------------------------------------------------------------------------------|--------------------------------------|
| Week 1          | Scope, objectives, roles, SRS, GitHub, contribution guidelines, AI-assisted README/task list | Requirements baseline and repository |
| Week 2          | AI-assisted wireframes, ERD, flowcharts, architecture, GitHub Projects                       | Design baseline                      |
| Weeks 3–4       | Authentication, navigation, basic UI, core structure, tests, Pull Requests                   | Working application skeleton         |
| Weeks 5–6       | CRUD, forms, API integration, AI-assisted tests, sprint monitoring                           | Functional core system               |
| Later           | Demand forecasting, supplier intelligence, Agentic AI, BI and approval workflow              | Integrated intelligent prototype     |
| Future versions | Logistics, financing partnerships, supplier network, industry agents                         | Scalable startup ecosystem           |

#### 16. AI-Assisted Development Policy

AI tools may assist with requirements, documentation, wireframes, code scaffolding, unit tests, debugging, task decomposition and progress summaries. All generated content/code must be reviewed, tested and adapted by the team. Secrets and private business data must not be shared with external AI tools.

#### 17. GitHub & Collaboration

- Repository will contain README, contribution guidelines, documentation and source code
- GitHub Issues/Projects will manage tasks and progress
- Feature branches will be used
- Pull Requests will support peer review
- Commit messages should clearly describe changes
- API keys, passwords, credentials and .env files must never be committed
- Weekly progress and retrospective information should be documented

#### 18. Version 1.0 Acceptance Criteria

- Users can register/login with role-based access
- Business can manage products, inventory, sales, suppliers and purchases
- Inventory updates correctly after transactions

- Owner can view meaningful KPIs
- System can produce a basic demand forecast
- System can compare suppliers and generate a procurement recommendation
- At least one end-to-end Agentic AI workflow can retrieve data, reason and prepare a controlled action
- Owner can approve/reject the proposed action
- Web and essential mobile interfaces work with the core system
- Project is documented and managed through GitHub

## 19. Assumptions and Constraints

- Version 1.0 is an academic/startup prototype, not a production financial platform
- AI accuracy depends on data quality and availability
- External LLM APIs may be used
- Time and team resources require feature prioritization
- High-impact AI actions require human approval
- Architecture should remain modular for future versions

## 20. Requirements Traceability to Course Plan

| Course Activity                 | SRS Coverage       | Evidence                            |
|---------------------------------|--------------------|-------------------------------------|
| Week 1: Scope/Objectives/Roles  | Sections 5–7       | SRS + README                        |
| Week 1: AI-assisted SRS         | Entire document    | SRS + AI-assistance note            |
| Week 1: GitHub/Contribution     | Section 17         | Repository + CONTRIBUTING.md        |
| Week 1: AI README/Task List     | Sections 15 and 17 | README + GitHub Issues/Projects     |
| Week 2: Wireframes/Architecture | Sections 12–14     | Wireframes + ERD + flowcharts       |
| Weeks 3–4: Core Development     | FR-01 to FR-05     | Source code + tests + Pull Requests |
| Weeks 5–6: CRUD/API             | FR-03 to FR-12     | Frontend/backend integration        |
| Later: Agentic AI               | FR-13 to FR-21     | Agent workflows + demonstration     |

## 21. Document Status

This document is the Version 1.0 requirements baseline. Requirements may be refined after teacher/team feedback and prototype testing. Major changes should be recorded as a new revision so that the project can evolve version by version.